

Distance-Time Graphs

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus On a distance-time graph, a flat line means stopped, a straight sloped line means constant speed, and a steeper slope means a faster speed.

Round 1 - Retrieve It

1. What does the horizontal axis usually represent?

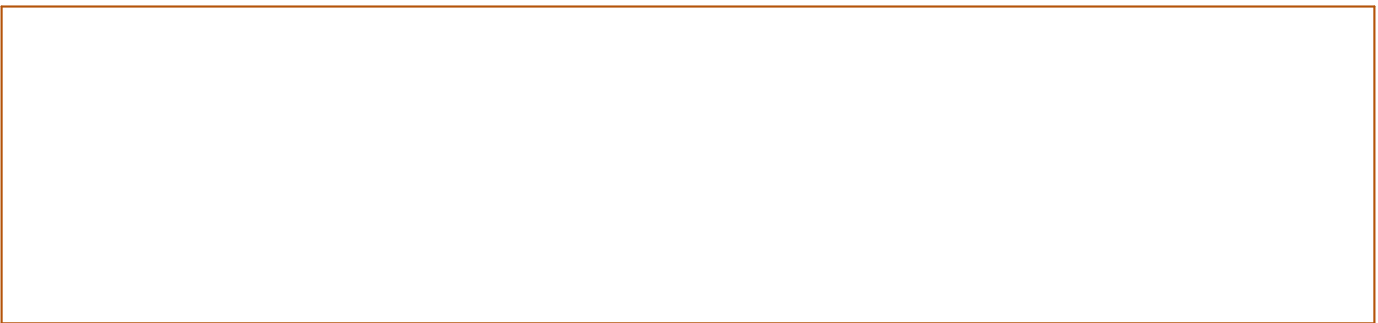
2. What does the vertical axis usually represent?

3. What does a flat segment mean?

4. How can you tell which of two straight segments represents faster motion?

Round 2 - Sketch It

Draw a distance-time graph for this trip: move steadily for 4 s, stop for 2 s, then move faster for 3 s. Label each segment.



Distance-Time Graphs - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. A line goes from 0 m at 0 s to 20 m at 4 s. Average speed?

2. The graph stays at 20 m from 4 s to 6 s. What is the object doing?

3. From 6 s to 10 s, distance increases from 20 m to 50 m. Average speed on this segment?

Round 4 - Reason from Evidence

Compare the first and last moving segments in the practice data. Which is faster, and how do both the numbers and the graph show it?

Distance-Time Graphs - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain Distance-Time Graphs in your own words. Use one example that is not printed on the cue card.

C**Connect**

Connect slope to speed. Why does a steeper distance-time line mean more distance is covered per unit time?

E**Extend**

Create a short story that matches a graph with a steep rise, a flat section, and then a gentle rise.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Distance-Time Graphs

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Time.
2. Distance from the starting point or distance traveled, depending on the graph.
3. Distance is not changing; the object is stopped.
4. The faster segment has the steeper slope.

Sketch It - look for

First rising straight line, then horizontal segment for 2 s, then a steeper rising line for 3 s.

Apply It

1. 5 m/s.
2. Stopped for 2 s.
3. $30 \text{ m} / 4 \text{ s} = 7.5 \text{ m/s}$.

Reason from Evidence - look for

Last segment is faster: 7.5 m/s vs 5 m/s, and its graph segment would be steeper.

ACE - Extend look-for

Accept stories that correctly map steep = fast, flat = stopped, gentle slope = slower constant motion.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.